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SECURE ONLINE VOTING SYSTEM USING CRYPTOGRAPHY

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**SECURE ONLINE VOTING SYSTEM USING CRYPTOGRAPHY**” has been carried out by **M. Nishanth (192511154), A. Meghanadh (192372281), N. Charan Sai (192372005), Ch. Mahivardhan (192411210)** under the supervision of **Dr. T. SANGEETHA** and is submitted in partial fulfilment of the requirements for the current semester of the B.E./B.Tech Computer Science and Engineering programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, **M. Nishanth (192511154), A. Meghanadh (192372281), N. Charan Sai (192372005Ch. Mahivardhan (192411210)**, of the Computer Science and Engineering Department, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**SECURE ONLINE VOTING SYSTEM USING CRYPTOGRAPHY**” is the result of our own Bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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INDIVIDUAL CONTRIBUTION STATEMENT

Mandatory for all team projects. Responsibilities are aligned with the four-module architecture used in the project.

Student / Reg. No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. %
N. Charan Sai 192372005	Module 1 – Voter Registration & Authentication	Designed voter enrolment, identity verification and login/session flow.	Verified registration validation rules and duplicate-vote prevention checks.	Chapter 1; Module 1 design, implementation and test-plan sections.	25%
Ch. Mahivardhan 192411210	Module 2 – RSA-Based Vote Encryption	Implemented RSA key generation and vote- ciphertext encryption logic.	Verified encryption/decryption correctness and key- size performance.	Chapter 2; Module 2 cryptographic design and methodology sections.	25%
M. Nishanth 192511154	Module 3 – Digital Signature Verification	Designed digital- signature generation and verification workflow.	Tested signature validity, tamper detection and rejection handling.	Chapter 3; Module 3 security, ethics and risk sections.	25%
A. Meghanadh 192372281	Module 4 – Vote Counting & Result Analysis	Designed the decryption- and-tally pipeline and result-analysis logic.	Verified tally accuracy and result- consistency checks.	Chapter 4–5; Module 4 results, conclusion and reflection sections.	25%
Total					100%

ABSTRACT

Traditional paper-based and simple electronic voting systems remain vulnerable to ballot tampering, impersonation, double voting, and a lack of independent verifiability, which undermines public trust in election outcomes. Online voting offers convenience and wider accessibility but introduces additional engineering challenges around confidentiality of the ballot, authentication of the voter, integrity of the transmitted vote, and transparent, auditable counting. This project addresses these challenges through a Secure Online Voting System built on established cryptographic primitives rather than relying on network security alone.

The proposed system is organised into four integrated modules. Module 1, Voter Registration and Authentication, enrolls eligible voters, verifies their identity, and issues credentials that prevent duplicate participation. Module 2, RSA-Based Vote Encryption, encrypts each ballot at the client side using RSA public-key cryptography so that the vote content is unreadable while in transit and in storage. Module 3, Digital Signature Verification, attaches a digital signature to every encrypted ballot so that the system can verify voter authenticity and detect any tampering before a vote is accepted for counting. Module 4, Vote Counting and Result Analysis, decrypts validated ballots using the authorised private key, aggregates them, and presents transparent, auditable results.

The system follows a register $\rightarrow$ encrypt $\rightarrow$ sign-and-verify $\rightarrow$ count workflow, ensuring that confidentiality, authenticity, and integrity are enforced at every stage rather than added afterward. The methodology combines RSA public-key encryption, SHA-based hashing, and digital-signature verification within a client–server architecture, with an encrypted database used as the vote store. The validation plan focuses on correctness of registration checks, encryption/decryption accuracy, signature verification and tamper detection, and tally correctness, with all numerical results to be reported only from actual implementation and testing. The project contributes a unified, module-wise cryptographic pipeline for online voting that keeps each security guarantee traceable to a specific stage of the system rather than treating encryption, authentication, and counting as isolated concerns.

Keywords: Online Voting, Cryptography, RSA Encryption, Digital Signature, Vote Integrity, Election Security

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LIST OF ABBREVIATIONS / SYMBOLS

Abbreviation	Description	Unit / Context
RSA	Rivest–Shamir–Adleman (public-key cryptosystem)	—
SHA-256	Secure Hash Algorithm, 256-bit	—
PKI	Public Key Infrastructure	—
OTP	One-Time Password	—
DB	Database	—
API	Application Programming Interface	—
UI	User Interface	—
TLS	Transport Layer Security	—
EVM	Electronic Voting Machine	—
bit	Key length unit for RSA modulus	bits
ms	Millisecond – used for encryption/verification timing	Time

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Voting is the primary mechanism through which citizens, members of an organisation, or shareholders express a collective decision, so the correctness and trustworthiness of the voting process is fundamental to the legitimacy of the outcome. Traditional paper-based ballots and stand-alone electronic voting machines require voters to be physically present, are logistically expensive to scale, and offer limited independent auditability once a ballot has been cast. As internet access and digital identity systems have become widespread, online voting has been proposed as a way to increase accessibility and reduce logistical overhead.

However, moving voting onto a network introduces new engineering risks: a vote transmitted or stored in plain form can be read, altered, or replayed by an attacker; a voter's identity can be spoofed; and a tally can be manipulated if there is no way to verify that a counted vote actually originated from an authenticated, unmodified ballot. This project addresses these risks directly through applied cryptography — specifically RSA public-key encryption for confidentiality and digital signatures for authenticity and integrity — rather than relying solely on network-level protections such as HTTPS.

1.2 Need for the Project

- Plaintext or weakly protected online ballots can be intercepted, read, or altered in transit.
- Without strong voter authentication, impersonation and duplicate voting become possible.
- A tally that cannot be independently verified against the cast ballots is vulnerable to silent manipulation.
- Centralised election servers are a high-value target, so confidentiality and integrity must be enforced cryptographically, not just administratively.
- An integrated registration – encryption – signature verification – counting pipeline can provide end-to-end assurance that is difficult to achieve with ad-hoc security measures.

1.3 Problem Statement

Existing simple online voting implementations largely depend on transport-layer security and server-side access control, but do not enforce end-to-end cryptographic confidentiality of the ballot content, verifiable authenticity of the voter for each individual vote, or tamper detection between the point of casting and the point of counting. The engineering challenge addressed by this project is to design a voter-registration, vote-encryption, signature-verification, and vote-counting pipeline that keeps the ballot confidential using RSA encryption, binds each ballot to an authenticated voter using a digital signature, detects tampering before a vote is accepted, and produces a verifiable, accurate tally — while remaining usable and computationally practical for a realistic number of voters.

1.4 Project Aim

To design and develop a secure online voting system that uses RSA-based encryption and digital signatures to guarantee voter authentication, ballot confidentiality, vote integrity, and accurate, auditable result computation.

1.5 Project Objectives

- To design a voter registration and authentication module that verifies voter eligibility and prevents duplicate registration or duplicate voting.
- To implement RSA public-key encryption so that each ballot is encrypted at the point of casting and remains unreadable in transit and storage.
- To design a digital-signature generation and verification mechanism that authenticates each voter's ballot and detects tampering.
- To implement a vote-counting and result-analysis module that decrypts only verified ballots and aggregates them into a transparent tally.
- To test the complete registration – encryption – signature verification – counting workflow using representative test votes and measurable acceptance criteria.
- To evaluate the system against confidentiality, integrity, authentication, and performance requirements.

1.6 Scope

Included:

- Voter registration, identity verification, and credential issuance.
- RSA key-pair generation and RSA-based encryption/decryption of vote data.
- Digital-signature generation, attachment, and verification for every submitted ballot.
- Vote decryption (of verified ballots only), tallying, and result presentation.
- Functional and security testing of the four integrated modules.

Excluded / Boundary Conditions:

- Nationwide biometric identity verification and government ID integration are outside the scope of the prototype; a simplified registration/authentication scheme is used instead.
- Blockchain-based distributed ledger implementation is not undertaken; the vote store is a centralised encrypted database, with distributed/ledger-based storage identified as future work.
- Physical polling-booth hardware and network infrastructure resilience (e.g., large-scale DDoS defence) are outside the scope of this software-focused project.
- Reported performance figures (encryption/decryption time, signature-verification time, tally accuracy) are limited to what is actually measured during testing; no results are assumed or invented.

1.7 Expected Engineering Outcome

The expected outcome is a working prototype of a secure online voting system comprising four integrated modules – voter registration and authentication, RSA-based vote encryption, digital-signature verification, and vote counting with result analysis. The prototype should demonstrate that a cast ballot remains confidential, that only ballots from authenticated voters with a valid, untampered signature are accepted for counting, and that the final tally accurately reflects the accepted ballots, together with a report of the system's measured security and performance characteristics.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review focused on recent research and existing systems related to electronic and online voting, public-key cryptography for ballot confidentiality, digital-signature-based authentication, blockchain-based voting, and homomorphic-encryption voting schemes. Sources were selected based on their relevance to voter authentication, ballot confidentiality, vote integrity, and end-to-end verifiability, and were used to identify the limitations that this project's engineering gap addresses.

2.2 Review of Existing Work

Approach	Technology / Method	Key Finding / Limitation
Traditional EVM-based voting	Stand-alone electronic voting machines with local tallying	Reduces manual counting error but requires physical presence and offers limited independent, cryptographic verifiability.
Web-based online voting over HTTPS	Transport Layer Security (TLS) with server-side access control	Protects data in transit but does not provide end-to-end ballot confidentiality or a voter-bound proof of ballot integrity once data reaches the server.
Blockchain-based voting systems	Distributed ledger, consensus protocols	Provides tamper-evident, distributed storage of votes but adds implementation complexity, scalability constraints, and depends on the security of the underlying network/consensus layer.
Homomorphic-encryption voting schemes	Homomorphic encryption enabling tallying on ciphertext	Allows counting without decrypting individual ballots, strengthening privacy, but is computationally expensive and complex to implement correctly.
Digital-signature-based e-voting research	Public-key cryptography with signature-based voter authentication	Provides strong authenticity and non-repudiation for cast ballots, but many implementations do not combine this with systematic ballot-confidentiality and tally-verification in a single evaluated pipeline.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
TLS-only web voting	Simple to deploy; protects transit data.	No end-to-end ballot confidentiality; server compromise exposes plaintext votes.	Motivates independent, application-level RSA encryption in Module 2.
Blockchain-based voting	Tamper-evident distributed record.	High implementation complexity; scalability and consensus overhead.	Informs future-work direction for distributed vote storage.
Homomorphic-encryption voting	Strong privacy; tallying without decrypting individual votes.	High computational cost; complex to implement and validate.	Considered as an alternative to RSA encryption in Section 3.4.
Digital-signature-based authentication research	Strong non-repudiation and tamper detection.	Often studied in isolation from confidentiality and tallying.	Basis for Module 3 and its integration with Modules 2 and 4.
Proposed four-module pipeline	Combines registration, RSA encryption, signature verification, and verifiable counting in one traceable workflow.	Prototype accuracy depends on key-management practice and test coverage.	Directly addresses the identified integration gap.

2.4 Research / Engineering Gap

The reviewed approaches show that transport-layer security, blockchain ledgers, homomorphic encryption, and digital signatures can each strengthen one aspect of online voting – transit protection, tamper-evident storage, tally privacy, or voter authenticity – but are frequently studied or deployed in isolation. A gap remains in combining voter registration and authentication, RSA-based ballot encryption, digital-signature-based integrity verification, and transparent counting into a single, module-wise pipeline where each security property is enforced at a clearly defined stage and can be independently tested. The proposed project addresses this integration gap using

cryptographic techniques (RSA and digital signatures) that are well understood, standardised, and practical to implement and validate within the scope of a capstone project.

2.5 Proposed Contribution

- A four-module, traceable pipeline: voter registration → RSA-based encryption → digital-signature verification → vote counting and result analysis.
- Application-level RSA encryption of the ballot content, independent of transport-layer security.
- Digital-signature verification applied to every ballot before it is eligible for counting, so tampered or unauthenticated ballots are rejected prior to the tally.
- A verifiable counting module that decrypts and aggregates only signature-verified ballots, keeping the tally traceable to accepted votes.
- An explicit test plan and security evaluation (confidentiality, integrity, authentication, performance) rather than an unvalidated claim of security.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Voter	Simple, trustworthy registration and voting process with confidentiality of their ballot.
Election administrator	Ability to configure an election, verify voter eligibility, and obtain an accurate, auditable tally.
System / security auditor	Verifiable evidence that ballots are authentic, unmodified, and counted correctly.
Project supervisor / evaluator	Traceable testing, measurable objectives, and transparent reporting of limitations.

Functional & Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	Register a voter and prevent duplicate registration.	Duplicate registration attempts rejected in all test cases.
Functional	Encrypt each ballot using RSA before transmission/storage.	No plaintext vote value present in transmitted or stored data.
Functional	Attach and verify a digital signature on every ballot.	Tampered or unsigned ballots rejected before counting.
Functional	Decrypt only verified ballots and compute the tally.	Tally count equals number of accepted (verified) ballots.
Non-Functional	Usability	Voter can complete registration and voting without inspecting raw cryptographic output.
Non-Functional	Reliability	Repeatable, identical decrypted output for identical valid input ciphertext.
Non-Functional	Performance overhead	Encryption, signing and verification time measured and reported per ballot.
Non-Functional	Security	Private keys are not exposed to the client; access to the tally function is restricted.

3.2 Constraints & Applicable Standards

The project is a software/cryptography-oriented system, so the constraints and standards most relevant are technical, data-security, and academic/ethical rather than physical or environmental.

Standard / Code	Requirement	How Applied in This Project
PKCS#1 / RSA cryptosystem specification	Defines correct RSA key generation, padding, and encryption/decryption procedure.	RSA key-pair generation and encryption/decryption implemented following the standard construction (public/private exponent, modulus size, padding).
FIPS 186 (Digital Signature Standard, referenced practice)	Defines accepted digital-signature generation and verification procedure.	Ballots are signed with the voter's private key and verified with the corresponding public key before acceptance.
NIST SP 800-series hashing guidance (referenced practice)	Recommends the use of a collision-resistant hash function (e.g., SHA-256) prior to signing.	SHA-256 is used to hash the ballot before signature generation and verification.
Information Technology Act, 2000 (India) – provisions on digital signatures and data protection	Establishes legal recognition of digital signatures and duties around data protection.	Referenced in the design of consent, credential handling, and signature-based authentication.
IEEE referencing practice	Sources should be consistently and traceably acknowledged.	IEEE-style references used for research papers, standards, software libraries, and datasets.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
RSA key size	2048-bit modulus (minimum)	bits	High
Hash function for signing	SHA-256	—	High
Vote encryption time	Measure and report per ballot	ms	Medium
Signature verification time	Measure and report per ballot	ms	Medium
Duplicate-vote prevention	0 duplicate ballots accepted per voter	count	High
Tally accuracy	Decrypted tally = number of accepted ballots	%	High
Tamper detection	100% rejection of ballots with invalid signature	%	High

3.4 Alternative Solutions & Evaluation

Alternative 1: TLS-only online voting. This approach relies entirely on transport-layer security to protect the ballot in transit, with plaintext or lightly protected storage on the server. It is simple to implement but leaves the vote exposed to anyone with server-side access.

Alternative 2: Homomorphic-encryption voting. This approach encrypts each ballot so that the tally can be computed directly on ciphertext without decrypting individual votes, offering strong per-vote privacy. It is more complex to implement correctly and computationally heavier for a capstone-scale prototype.

Alternative 3 (Proposed): RSA encryption with digital-signature verification. Each ballot is individually encrypted with RSA and signed by the voter; the server verifies the signature before decrypting and counting. This balances strong, well-understood security guarantees with practical implementation effort.

Criterion	Weight	TLS-Only Voting	Homomorphic-Encryption Voting	RSA + Digital Signature (Proposed)
Ballot confidentiality	High	Low	High	High
Voter authenticity / non-repudiation	High	Low	Medium	High
Implementation complexity	Medium	Low	High	Medium
Computational cost	Medium	Low	High	Medium
Auditability of tally	High	Low	Medium	High
Project relevance	High	Low	Medium	High

3.5 Selected Design & Detailed Design

The RSA-with-digital-signature design is selected because it directly satisfies the project's confidentiality, authenticity, and integrity requirements with well-established, standardised cryptographic primitives that are practical to implement, test, and validate within the project timeline, unlike the higher-complexity homomorphic approach, and unlike TLS-only voting it does not depend solely on the security of the transport channel or the server operator.

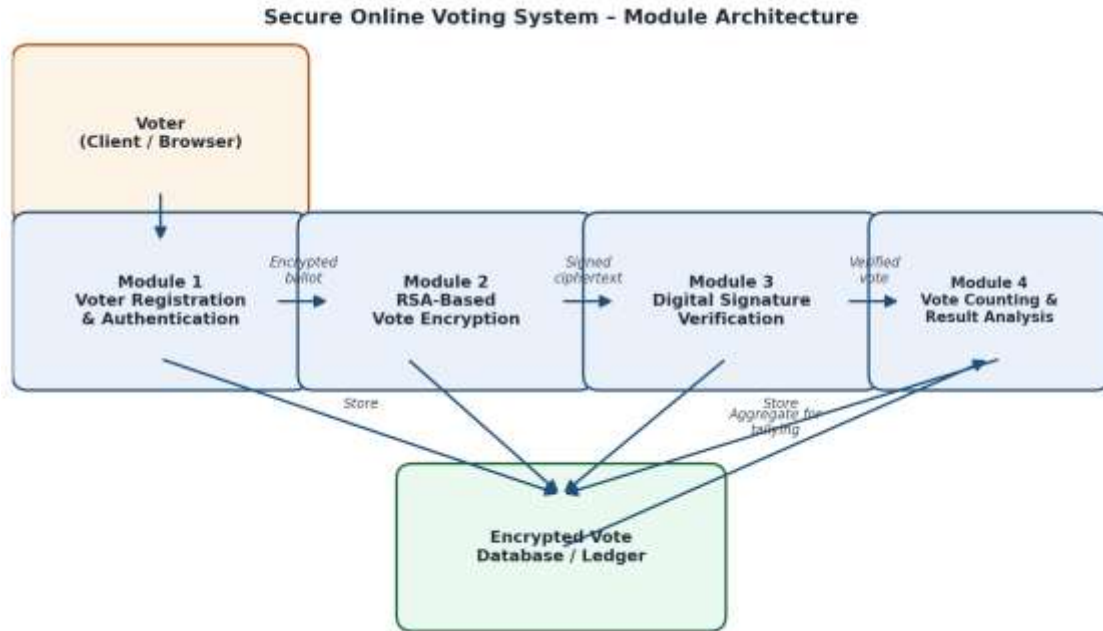


Figure 1. Proposed System Architecture

Systems Approach

Module 1 authenticates the voter and issues a registration record that Module 2 relies on before accepting a ballot for encryption. Module 2 encrypts the ballot with RSA and hands the ciphertext to Module 3, which attaches and later verifies a digital signature over the hashed ciphertext. Only ballots that pass signature verification are forwarded to Module 4, which decrypts them with the authorised private key, aggregates the results, and produces the final, auditable tally. Each module therefore depends on a clearly defined output from the previous module, keeping responsibilities – and testing – separated.

Working Principle

- Step 1 – Registration: voter provides identity details and credentials; the system checks eligibility and prevents duplicate registration.
- Step 2 – Key issuance: an RSA key pair is associated with the election (and/or the voter, depending on the chosen scheme); the public key is distributed for encryption.
- Step 3 – Vote encryption: the voter's ballot selection is encrypted at the client using the RSA public key before transmission.
- Step 4 – Hashing and signing: the ciphertext is hashed (SHA-256) and signed with the voter's private signing key.

- Step 5 – Signature verification: the server verifies the signature against the voter's registered public key; ballots that fail verification are rejected.
- Step 6 – Decryption and tally: verified ciphertexts are decrypted using the election's private key and aggregated into the final result.
- Step 7 – Result presentation: the aggregated, verifiable tally is displayed / reported.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Vote encryption scheme	TLS-only protection	Simple to deploy.	No confidentiality once data reaches the server.	RSA application-level encryption selected to protect the ballot end-to-end.
Authenticity mechanism	Password-only authentication	Familiar, low implementation effort.	Cannot prove ballot integrity or bind a specific ballot to a voter.	Digital signature selected for non-repudiation and tamper detection.
Four-module architecture	Single monolithic voting script	Faster initial development.	Harder to test and assign responsibility across the team.	Four-module design selected for clear separation of duties and independent testing.
Counting approach	Decrypt-all-then-filter	Simpler control flow.	Risks tallying unverified/tampered ballots if verification is skipped.	Verify-then-decrypt selected so only authenticated, untampered ballots are ever counted.

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Cryptographic operations should not impose unnecessary computational/energy overhead.	Measured encryption/signing/verification time per ballot.	RSA with a standard 2048-bit key selected instead of a heavier scheme to keep per-vote overhead practical.
Ethics	Voter data and results must be handled transparently and honestly, without fabricated results.	Test records, registration logs, and reported limitations.	Only actually measured performance and tally figures are reported; assumptions are stated explicitly.
Health & Safety	The project is a software system and introduces no physical hazard.	Development and testing environment review.	No physical safety mitigation required beyond standard workstation practice.
Security	Private keys and voter credentials must not be exposed or logged in plaintext.	Review of key storage, access control, and logging design.	Private keys restricted to the signing/decrypting party; only

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
			ciphertext and signatures are transmitted or logged.

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Private key compromise	Low	High	High	Restrict key storage/access; do not transmit private keys.	Medium
Duplicate voting / registration bypass	Medium	High	High	Enforce unique-voter checks at registration and before tally acceptance.	Medium
Ballot tampering in transit	Medium	High	High	Digital-signature verification rejects any modified ciphertext.	Low
Weak or reused RSA keys	Low	High	Medium	Generate fresh, sufficiently large (≥ 2048 -bit) keys per election.	Low
Performance overhead at scale	Medium	Medium	Medium	Measure per-ballot timing during testing and report scalability limits.	Medium
Incorrect tally due to software defect	Low	High	Medium	Unit-test decryption and aggregation logic against known test ballots.	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project follows an incremental, module-wise development approach. The voter registration and authentication module is implemented first, followed by RSA key generation and vote encryption, then digital-signature generation and verification, and finally the decryption, tallying, and result-analysis module. Each module is unit-tested individually before the four modules are integrated and tested end-to-end using representative test ballots.

Resource	Purpose
Development system	Implementation, testing, and documentation of all four modules.
Cryptographic library (e.g., RSA/SHA implementation)	Key generation, encryption/decryption, hashing, and signing/verification.
Test voter/ballot dataset	Functional and security testing of registration, encryption, and tally logic.
Database / vote store	Storage of registration records and encrypted, signed ballots.
Version control repository	Code management and team collaboration across the four modules.

4.2 Software Implementation & Modern Tools Used

Tool / Software	Purpose	Stage Used
Python / Java (implementation language)	Implementation of registration, RSA encryption, signature logic and tally computation.	Development and testing
Cryptography library (RSA, SHA-256)	Key-pair generation, ballot encryption/decryption, hashing, digital signing and verification.	Module 2 & 3 implementation
Relational / encrypted database	Storage of voter registration records and encrypted, signed ballots.	Modules 1, 2 and 4
Web / application framework	Voter-facing registration and voting interface.	Integration
Git / GitHub	Version control and team collaboration.	Throughout development
Dashboard / reporting interface	Presentation of tally and result analysis.	Module 4 / presentation

4.3 Test Plan & Verification

The following test plan is intended to be completed against the final implementation. Achieved values must be filled in from actual execution and are not to be invented in advance.

Requirement	Test Method	Acceptance Criterion	Achieved Value	Status
Voter registration & duplicate prevention	Register the same voter twice; register with invalid eligibility data.	Second/invalid registration attempt rejected.	To be measured	Pending
RSA encryption correctness	Encrypt a known ballot value and decrypt with the corresponding private key.	Decrypted value exactly matches original ballot value.	To be measured	Pending
Digital-signature verification	Submit a ballot with a valid signature and a ballot with a modified ciphertext/invalid signature.	Valid signature accepted; tampered/invalid signature rejected.	To be measured	Pending
Tally accuracy	Cast a known set of test ballots and compare computed tally with expected counts.	Computed tally matches expected counts exactly.	To be measured	Pending
Encryption/verification performance	Measure time taken per ballot for encryption, signing and verification.	Timings recorded and reported for the tested key size.	To be measured	Pending

4.4 Results

Results are to be inserted here as measured tables, graphs, or simulation output once the implementation and test plan in Section 4.3 have been executed — for example, encryption/decryption timing versus key size, signature-verification success/rejection counts, and the final test-election tally. No numerical results are presented in this report in advance of actual measurement.

4.5 Data Analysis & Engineering Judgment

Once results are available, this section should interpret them against the acceptance criteria in Section 4.3 — for example, whether encryption/verification timing scales acceptably with the number of voters, whether any tampered ballots were incorrectly accepted (a critical security failure to report honestly if observed), and whether the tally exactly matches the expected count. Any deviation from expected behaviour should be explained together with its engineering implication and any corrective change made to the design.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully / Partially / Not Achieved)
Design voter registration & duplicate prevention	Registration test results (Section 4.3)	To be assessed after testing
Implement RSA-based vote encryption	Encryption/decryption correctness tests	To be assessed after testing
Implement digital-signature verification	Signature acceptance/rejection tests	To be assessed after testing
Implement vote counting & result analysis	Tally accuracy test	To be assessed after testing
Evaluate performance and security	Timing measurements and security test cases	To be assessed after testing

4.7 Strengths & Limitations

Strengths:

- Ballot confidentiality is enforced end-to-end using RSA encryption rather than relying only on transport security.
- Digital-signature verification provides tamper detection and non-repudiation for every submitted ballot.
- The four-module design keeps registration, encryption, verification, and counting independently testable.

Limitations:

- The prototype uses a simplified identity-verification scheme rather than a full government-grade identity system.
- Vote storage is centralised rather than distributed/ledger-based, which is identified as future work.
- RSA encryption/signing performance may need further optimisation for very large voter populations.

4.8 Project Planning, Roles & Budget

Student	Assigned Responsibility	Actual Contribution
N. Charan Sai (192372005)	Module 1 – Voter Registration & Authentication	To be confirmed on completion
Ch. Mahivardhan (192411210)	Module 2 – RSA-Based Vote Encryption	To be confirmed on completion
M. Nishanth (192511154)	Module 3 – Digital Signature Verification	To be confirmed on completion
A. Meghanadh (192372281)	Module 4 – Vote Counting & Result Analysis	To be confirmed on completion

Item	Estimated Cost	Actual Cost
Development / cloud environment	[₹ amount]	[₹ amount]
Domain / hosting (if applicable)	[₹ amount]	[₹ amount]
Miscellaneous (testing, documentation)	[₹ amount]	[₹ amount]
Total	[₹ amount]	[₹ amount]

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This project set out to design and develop a Secure Online Voting System using cryptography, organised into four integrated modules: Voter Registration and Authentication, RSA-Based Vote Encryption, Digital Signature Verification, and Vote Counting and Result Analysis. The design demonstrates how RSA public-key encryption and digital signatures can be combined into a single, traceable register → encrypt → sign-and-verify → count pipeline so that confidentiality, authenticity, and integrity are enforced cryptographically at each stage rather than left to network security or administrative controls alone. Once the test plan in Chapter 4 is executed, its results will validate whether the implementation meets the stated functional and security requirements; the design and evaluation framework presented in this report provide the basis for that validation.

5.2 Future Work

- Replace the centralised vote store with a distributed, tamper-evident ledger (e.g., blockchain-based storage) for stronger auditability.
- Integrate stronger, government-grade identity verification (e.g., biometric or Aadhaar-based e-KYC) for voter registration.
- Explore homomorphic encryption or mix-net techniques to allow tallying without ever decrypting individual ballots.
- Conduct larger-scale performance testing to evaluate the system's behaviour under a realistic number of concurrent voters.
- Add a voter-facing verifiability mechanism so each voter can confirm their own ballot was correctly counted without revealing their choice.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Practical application of RSA public-key cryptography for confidentiality in a real system context.
- Digital-signature generation and verification as a mechanism for authenticity and tamper detection.

- Secure system design practices, including threat/risk identification and mitigation for a security-critical application.

Engineering & Professional Skills Developed:

- Module-wise system design and integration across a four-person team.
- Structured test planning and honest, evidence-based reporting of results.
- Trade-off analysis and justified engineering decision-making under security, performance, and time constraints.

Individual Reflection (each student, 4–5 lines):

M. Nishanth (192511154) – Module 3 – Digital Signature Verification

Working on Module 3 improved my understanding of digital signatures, RSA cryptography, hashing, and signature verification in a secure online voting system. The most difficult engineering problem was ensuring that a vote and its associated digital signature could be verified correctly while detecting any modification to the vote data. I addressed this by using a secure hash of the vote data and verifying the signature with the corresponding public key before accepting the vote. I independently learned how private and public keys provide authentication, integrity, and non-repudiation in digital communication. If the project were repeated, I would improve the module by adding stronger key-management mechanisms and more detailed verification logs for security auditing.

REFERENCES

References are to be listed in a consistent, recognised style (IEEE recommended). All literature, standards, websites, datasets, software/libraries, and AI-assisted resources used shall be acknowledged. Sample entries (to be replaced/expanded with the sources actually used):

- [1] R. L. Rivest, A. Shamir, and L. Adleman, “A Method for Obtaining Digital Signatures and Public-Key Cryptosystems,” Communications of the ACM, vol. 21, no. 2, pp. 120–126, 1978.
- [2] National Institute of Standards and Technology, “Digital Signature Standard (DSS),” FIPS PUB 186-5, 2023.
- [3] National Institute of Standards and Technology, “Secure Hash Standard (SHS),” FIPS PUB 180-4, 2015.
- [4] RSA Laboratories, “PKCS #1: RSA Cryptography Specifications,” RFC 8017, 2016.
- [5] Government of India, “Information Technology Act, 2000,” [Online resource, as applicable].
- [6] [Add relevant research papers on electronic/online voting, blockchain voting, or homomorphic-encryption voting used during the literature review.]
- [7] [Add documentation for the cryptography library / framework used in implementation.]

APPENDICES

Appendix A – Detailed Calculations

RSA key-generation and encryption/decryption worked example.

Appendix B – Engineering Drawings / Schematics

Detailed sequence diagram of registration, encryption, signing and verification.

Appendix C – Source Code / Code Repository Information

Repository link and essential code excerpts only; full source maintained separately.

Appendix D – Datasheets

Not applicable – software-only project (omit or replace if hardware is used).

Appendix E – Experimental / Raw Data

Raw test-ballot data and timing measurements once testing is completed.

Appendix F – Ethics / Institutional Approval

Where applicable.

Appendix G – Project Meeting / Progress Records

Team meeting logs and progress notes.

Appendix H – User / Industry Feedback

Where applicable.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Where applicable.

Appendix J – Individual Contribution Evidence

Mandatory for team projects – supporting evidence for Section “Individual Contribution Statement.”

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department / faculty assessor.

Outcome	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written / oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal / environmental impact	SO2 / SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2 / SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated / system-level engineering	SO1 / SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)